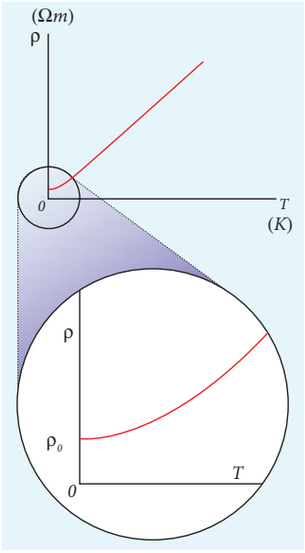


Figure 2.13 (a) Temperature dependence of resistivity for a conductor (b) Non linear region at low temperature

The temperature coefficient can be also be obtained from the equation (2.28),

$$R_T - R_\circ = \alpha R_\circ \left(T - T_\circ\right)$$

$$\therefore \alpha = \frac{R_T - R_\circ}{R_\circ \left(T - T_\circ\right)} = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T}$$

$$\alpha = \frac{1}{R_\circ}\frac{\Delta R}{\Delta T} \qquad (2.29)$$

where $\Delta R = R_T - R_\circ$ is change in resistance during the change in temperature $\Delta T = T - T_\circ$

α of semiconductors

For semiconductors, the resistivity decreases with increase in temperature. As the temperature increases, more electrons will be liberated from their atoms (Refer unit 9 for conduction in semi conductors). Hence the current increases and therefore the resistivity decreases as shown in Figure 2.14. A semiconductor with a negative temperature coefficient of resistance is called a thermistor.

The typical values of temperature coefficients of various materials are given in table 2.3.

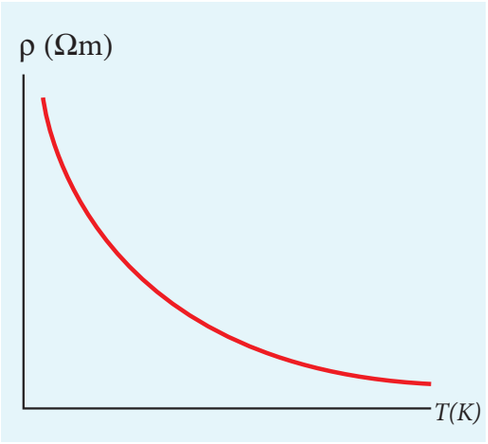


Figure 2.14 Temperature dependence of resistivity for a semiconductor

Table 2.3

Color	Temperature Coefficient α [(°C)$^{-1}$]
Silver	3.8×10^{-3}
Copper	3.9×10^{-3}
Gold	3.4×10^{-3}
Aluminum	3.9×10^{-3}
Tungsten	4.5×10^{-3}
Iron	5.0×10^{-3}
Platinum	3.92×10^{-3}
Lead	3.9×10^{-3}
Nichrome	0.4×10^{-3}
Carbon	-0.5×10^{-3}
Germanium	-48×10^{-3}
Silicon	-75×10^{-3}

We can understand the temperature dependence of resistivity in the following way. In section 2.1.3, we have shown that the electrical conductivity, $\sigma = \frac{ne^2\tau}{m}$. As the resistivity is inverse of σ, it can be written as,

$$\rho = \frac{m}{ne^2\tau} \qquad (2.30)$$